DATA SET DS1406

10. Marriage Rates Half of adults today are married compared with the all-time high of 72% in 1960.[9] Do marriage rates vary by educational level? A sample of 300 adults in each of three educational categories provided the following information.

	Married		
Educational Level	**Yes**	**No**	
Bachelors or higher	187	113	300
Some college	162	138	300
High school or less	149	151	300
	498	402	900

Test to determine whether marriage rates differ significantly among adults in these educational levels using $\alpha = .01$. How would you summarize your results?

Ho : marriage rates and educational levels independent.

Ha : marriage rates and educational levels not independent.

Yes ∩ $E = \frac{300 \times 498}{900} = 166$

No ∩ $E = \frac{300 \times 402}{900} = 134$

Bachelors or higher

Yes ∩ $\chi^2 = \frac{(187-166)^2}{166} \approx 2.6566$

No ∩ $\chi^2 = \frac{(113-134)^2}{134} \approx 3.2910$

Some college

Yes ∩ $\chi^2 = \frac{(162-166)^2}{166} \approx 0.0964$

No ∩ $\chi^2 = \frac{(138-134)^2}{134} \approx 0.1194$

High school or less

Yes ∩ $\chi^2 = \frac{(149-166)^2}{166} \approx 1.7410$

No ∩ $\chi^2 = \frac{(151-134)^2}{134} \approx 2.1567$

Total $\chi^2 = 2.6566 + 3.2910 + \cdots$

$= 10.061$

$df : (3-1)(2-1) = 2$

$\chi^2_{0.01,2} = 9.210$

$\therefore 10.06 > 9.210 \in RR$

→ Reject Ho

DATA SET DS1407

12. How Big Is the Household? A local chamber of commerce surveyed 120 households in their city—40 in each of three types of residence (apartment, duplex, or single residence)—and recorded the number of family members in each of the households. The data are shown in the table.

	Type of Residence			
Family Members	**Apartment**	**Duplex**	**Single Residence**	
1	8	20	1	29
2	16	8	9	33
3	10	10	14	34
4 or More	6	2	16	24
	40	40	40	120

Is there a significant difference in the family size distributions for the three types of residence? Test using $\alpha = .01$. If there are significant differences, describe the nature of these differences.

Ho : 3种 residence ∩ family members 无 significantly different.

Ha : " 有 significantly different.

1 members ∩ $E : \frac{29 \times 40}{120} = 9.667$

2 " $E : \frac{33 \times 40}{120} = 11$

3 " $E : \frac{34 \times 40}{120} = 11.33$

4 or more $E : \frac{24 \times 40}{120} = 8.000$

1 member

apartment : $\frac{(8-9.667)^2}{9.667} \approx 0.287$

Duplex : $\frac{(20-9.667)^2}{9.667} \approx 11.046$

Single Resident : $\frac{(1-9.667)^2}{9.667} \approx 7.771$

2 member

apartment : $\frac{(16-11)^2}{11} \approx 2.273$

Duplex : $\frac{(8-11)^2}{11} \approx 0.818$

Single Resident : $\frac{(9-11)^2}{11} \approx 0.364$

3 member

apartment : $\frac{(10-11.33)^2}{11.33} \approx 0.157$

Duplex : $\frac{(10-11.33)^2}{11.33} \approx 0.157$

Single Resident : $\frac{(14-11.33)^2}{11.33} \approx 0.628$

4 or more

apartment : $\frac{(6-8)^2}{8} = 0.5$

Duplex : $\frac{(2-8)^2}{8} = 4.5$

Single Resident : $\frac{(16-8)^2}{8} = 8$

$\chi^2 = 0.287 + 11.046 + 7.771 \cdots$

≈ 36.50

$df = (4-1)(3-1) = 6$

$\chi^2_{0.01,6} = 16.812$

$\therefore 36.50 > 16.812 \in RR$

→ Reject Ho.

12. Dissolved Oxygen Content Dissolved oxygen content is a measure of the ability of a river, lake, or stream to support aquatic life, with high levels being better. A pollution-control inspector who suspected that a river community was releasing semitreated sewage into a river, randomly selected five specimens of river water at a location above the town and another five below. These are the dissolved oxygen readings (in parts per million):

Above Town	4.8	5.2	5.0	4.9	5.1
Below Town	5.0	4.7	4.9	4.8	4.9

a. Use a one-tailed Wilcoxon rank sum test with $\alpha = .05$ to confirm or refute the theory.

b. Use a Student's t-test (with $\alpha = .05$) to analyze the data. Compare the conclusion reached in part a.

a.

数值	组别	Ranking 計算	最终排名
4.7	Below	第1名	1
4.8	Above	2,3名平均 $\to \frac{2+3}{2}$	2.5
4.8	Below		2.5
4.9	Above	第4,5,6名平均 $\to \frac{4+5+6}{3}$	5
4.9	Below		5
4.9	Below		5
5.0	Above	第7,8名平分 $\to \frac{7+8}{2}$	7.5
5.0	Below		7.5
5.1	Above	第9名	9
5.2	Above	第10名	10

Below a rank sum $= 1 + 2.5 + \cdots = 21$

Above a rank sum $= 2.5 + 5 \cdots = 34$

$\alpha = 0.05,\ n_1 = 5,\ n_2 = 5$

$T_L = 19$

$\therefore\ 21 > 19,\ \notin RR$

$\longrightarrow$ Fail to reject H_0.

b. Above, 1

$$\bar{x}_1 = \frac{4.8 + 5.2 + 5.0 + 4.9 + 5.1}{5} = 5$$

$$\Sigma x_1^2 = 4.8^2 + 5.2^2 + 5.0^2 + 4.9^2 + 5.1^2 = 125.1$$

$$s_1^2 = \frac{125.1 - 5 \times 5.0^2}{5-1} = \frac{0.1}{4} = 0.025$$

Below, 2

$$\bar{x}_2 = \frac{5.0 + 4.7 + 4.9 + 4.8 + 4.9}{5} = 4.86$$

$$\Sigma x_2^2 = 5.0^2 + 4.7^2 + 4.9^2 + 4.8^2 + 4.9^2 = 118.15$$

$$s_2^2 = \frac{118.15 - 5 \times 4.86^2}{5-1} = \frac{0.052}{4} = 0.013$$

$$S_p^2 = \frac{(5-1) \times 0.025 + (5-1) \times 0.013}{5 + 5 - 2} = 0.019$$

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{S_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} = \frac{5 - 4.86}{\sqrt{0.019 \left(\frac{1}{5} + \frac{1}{5}\right)}} = \frac{0.14}{\sqrt{0.0076}} = 1.606$$

$df = n_1 + n_2 - 2 = 8$

$t_{0.05, 8} = 1.860$

$\therefore\ 1.606 < 1.860 \longrightarrow$ fail to reject H_0.

$\longrightarrow$ same with (a)

DATA SET DS1502

13. Eye Movement In an investigation of the visual scanning behavior of deaf children, measurements of eye movement were taken on nine deaf and nine hearing children. The table gives the eye-movement rates and their ranks (in parentheses). Does it appear that the distributions of eye-movement rates for deaf children and hearing children differ?

	Deaf Children	Hearing Children
	2.75 (15)	.89 (1)
	2.14 (11)	1.43 (7)
	3.23 (18)	1.06 (4)
	2.07 (10)	1.01 (3)
	2.49 (14)	.94 (2)
	2.18 (12)	1.79 (8)
	3.16 (17)	1.12 (5.5)
	2.93 (16)	2.01 (9)
	2.20 (13)	1.12 (5.5)
Rank Sum	126	45

H_0 : Deaf children and hearing children eye-movement 无 significant different.

H_a : " 有 "

$n_1 = 9,\ n_2 = 9,\ N = 18,\ \alpha = 0.05$

$W_1 = 126$

$W_2 = 45$

when $\alpha = 0.05,\ n_1 = 9,\ n_2 = 9$

$T_L = 62 \quad (T_U = 9 \times (9 + 9 + 1) - 62 = 109)$

$W = 45 < 62$ and $126 > 109 \in RR$

$\longrightarrow$ Reject H_0.

Two Simple Examples *Use the sign test to compare two populations for significant differences for the paired data in Exercises 4–5. State the null and alternative hypotheses to be tested. Determine an appropriate rejection region with $\alpha \le .10$. Calculate the observed value of the test statistic and present your conclusion.*

4.

	Pair						
Population	**1**	**2**	**3**	**4**	**5**	**6**	**7**
1	8.9	8.1	9.3	7.7	10.4	8.3	7.4
2	8.8	7.4	9.0	7.8	9.9	8.1	6.9

H_0 : 2个母体分佈无明显差異 $(p=0.5)$

H_a : 〃 有明显差異 $(p\neq 0.5)$

Pair 1 : $8.9-8.8=+0.1 \to (+)$

〃 2 : $8.1-7.4=+0.7 \to (+)$

〃 3 : $9.3-9.0=+0.3 \to (+)$

〃 4 : $7.7-7.8=-0.1 \to (-)$

〃 5 : $10.4-9.9=+0.5 \to (+)$

〃 6 : $8.3-8.1=+0.2 \to (+)$

〃 7 : $7.4-6.9=+0.5 \to (+)$

$X \sim B(7, 0.5)$, 要求 $\alpha \le 0.10$

$P(X\le 0) = 0.5^7 = 0.0078$

$P(X\le 1) = P(X=0)+P(X=1) = 0.0078 + 7\times 0.5^7 = 0.0625$

$\therefore P(X\le 1) = 0.0625 > 0.05$ (two tail $\to$ $0.0625\times 2 = 0.125 > 0.10$)

$X=0$ or $X=7$

$\alpha = P(X=0)+P(X=7) = 0.0078\times 2 = 0.0156 < 0.10$

$X=6 \notin RR$

$\to$ fail to reject H_0.

11. Gourmet Cooking Two chefs, A and B, rated 22 meals on a scale of 1 – 10. The data are shown in the table. Do the data provide sufficient evidence to indicate that one of the chefs tends to give higher ratings than the other? Test by using the sign test with a value of α near .05.

DATA SET DS1507

Meal	A	B		Meal	A	B	
1	6	8	-	12	8	5	+
2	4	5	-	13	4	2	+
3	7	4	+	14	3	3	=
4	8	7	+	15	6	8	-
5	2	3	-	16	9	10	-
6	7	4	+	17	9	8	+
7	9	9	=	18	4	6	-
8	7	8	-	19	4	3	+
9	2	5	-	20	5	4	+
10	4	3	+	21	3	2	+
11	6	9	-	22	5	3	+

a. Use the binomial tables in Appendix I to find the exact rejection region for the test.

b. Use the large-sample z statistic. (NOTE: Although the large-sample approximation is suggested for $n \ge 25$, it works fairly well for values of n as small as 15.)

c. Compare the results of parts a and b.

(c) (a) and (b) have the same result.

$n = 22-2 = 20$, $+ = 11$, $- = 9$, $X = 11$, $\alpha = 0.05$

(a) 在 H_0 $(p=0.5)$ 成立下, $X \sim B(20, 0.5)$

$P(X\le 5) = 0.0207$

$P(X\le 6) = 0.0577 \to$ 超 0.025

choose $X\le 5$ 为 left tail rr, right tail $X \le 20-5 = 15 \to X \ge 15$, $\alpha = 0.0207\times 2 = 0.0414$

$X=11 \notin RR$

$\to$ fail to reject H_0

(b) $\mu = n\times p = 20\times 0.5 = 10$

$\sigma = \sqrt{n\times p\times(1-p)} = \sqrt{20\times 0.5\times 0.5} = 2.236$

$$z = \frac{(x-0.5)-\mu}{6} = \frac{(11-0.5)-10}{2.236} = \frac{0.5}{2.236} = 0.224$$

$\therefore 0.224 < 1.96 \notin RR$

$\to$ Fail to reject H_0.

DATA SET DS1508

13. Recovery Rates Clinical data concerning the effectiveness of two drugs in treating a particular condition (as measured by recovery in 7 days or less) were collected from 10 hospitals. You want to know whether the data present sufficient evidence to indicate a higher recovery rate for one of the two drugs.

a. Test using the sign test. Choose your rejection region so that α is near .05.

b. Why might it be inappropriate to use the Student's t-test in analyzing the data?

	Drug A			
Hospital	Number in Group	Number Recovered (7 days or less)	Percentage Recovered	
1	84	63	75.0	-
2	63	44	69.8	-
3	56	48	85.7	+
4	77	57	74.0	-
5	29	20	69.0	-
6	48	40	83.3	+
7	61	42	68.9	-
8	45	35	77.8	-
9	79	57	72.2	-
10	62	48	77.4	-

(+) = 2
(-) = 8
$n = 10$
$X = 2$

	Drug B		
Hospital	Number in Group	Number Recovered (7 days or less)	Percentage Recovered
1	96	82	85.4
2	83	69	83.1
3	91	73	80.2
4	47	35	74.5
5	60	42	70.0
6	27	22	81.5
7	69	52	75.4
8	72	57	79.2
9	89	76	85.4
10	46	37	80.4

(a) H_0 : 2种药物の康復率无 significant different ($p=0.5$)
H_a : 〃 有 〃 ($p \neq 0.5$)

$\alpha = 0.05$

$X \sim B(10, 0.5)$

$P(X \leq 1) = 0.0107$

$P(X \leq 2) = 0.0547$ ($0.0547 \times 2 = 0.1094 \longrightarrow$ 离 0.05 太远)

α 接近 0.05，选 $X \leq 1$ 为 left tail，right tail 为 $X \geq 10-1 = 9$

$RR = \{X \leq 1 \text{ or } X \geq 9\}$

$\alpha = 0.0107 \times 2 = 0.0214 \longrightarrow$ 最接近 0.05.

$X = 2$, $\notin RR$

$\longrightarrow$ fail to reject H_0.

(b) 1. Normality Assumption 遭质疑
2. Variability & Weighting.